Bill of Materials and Design Rationale: 1.2 m Electric High-Wing RC Aircraft

1. Aircraft Concept and Assumptions

The aircraft is a four-channel, high-wing, tractor-motor trainer-style plane with a conventional tail, built to be light, strong and quick to repair.

Parameter	Value (assumed)	Why this value
Wingspan	1.2 m	Top of the allowed range. More span at the same chord gives more wing area, so lower wing loading and slower, gentler flight. The cost is a bigger airframe to transport.
Chord / wing area	0.22 m / about 0.26 m²	Wide enough to keep the wing stiff and the Reynolds number near 1.7×10^5 (11 m/s × 0.22 m / 1.46×10^{-5}).
Aspect ratio	about 5.5 (span² / area)	Moderate. It is efficient enough but stays stiff and crash-tolerant. Very long thin wings are fragile.
Layout	High wing, conventional tail, nose-mounted motor	A high wing makes the plane self-righting (the weight hangs below the wing), keeps the propeller well clear of the ground, and lets the wing be held on with rubber bands that release in a crash.
Wing section	Flat-bottom or mild camber, about 12% thick (NACA 4412 or Clark Y type)	Your XFLR5 run for NACA 4412 at Re 250,000 gave CLmax of at least 1.44, a gradual stall and a mild nose-down moment (Cm −0.107), which makes trimming easy. A flat bottom is also easy to build. Our Re is a little lower than 250,000, so expect slightly less lift.

Parameter	Value (assumed)	Why this value
Control	4 channels: throttle, aileron, elevator, rudder	Full three-axis control without complexity.
Takeoff mass	about 0.95 kg (Section 5)	Built up from the part masses with a 10% allowance.
Wing loading	about 3.6 kg/m²	Low, so the plane flies slowly and can be hand-launched.
Stall / cruise speed	about 7 m/s / 11 to 14 m/s	From CLmax of 1.2 for the real wing (assumed).
Target flight time	about 10 min	Enough for a competition flight with margin.

Other assumptions: sea-level air (density 1.225 kg/m³); 3-cell (3S) LiPo power; ground or hand launch from short grass; prices and weights are estimates to be replaced with real product data.

2. Bill of Materials (Parts A and B)

Recommendation 1 is the budget-oriented choice and Recommendation 2 is the performance-oriented choice. Costs are ₹ for the whole line, shown as Rec 1 / Rec 2.

Component	Recommendation 1 (budget)	Recommendation 2 (performance)	Key specifications	Qty	Cost ₹
Airframe sheets	5 mm foam board (sun board)	Balsa sheets (1.5, 3, 6 mm) + 3 mm light plywood + heat-shrink film	Rec 1 is stiff, cheap and dent-resistant. Rec 2 gives a structure about 25 to 30% lighter (assumed) and a better surface.	4 sheets / 1 set	480 / 3,000
Wing spar and reinforcement	Carbon-fibre tube 8 × 6 mm, about 1.1 m	Same tube plus glass-fibre tape at the wing root	Sized by the bending check in Section 3.5	1	450 / 600

Component	Recommendation 1 (budget)	Recommendation 2 (performance)	Key specifications	Qty	Cost ₹
Motor	Generic A2212 1000 KV outrunner	SunnySky X2212 about 980 KV class	KV about 1000; max current 12 to 15 A (typical); mass 50 to 55 g; 3S; 10×4.7 prop; shaft 3.17 to 5 mm	1	650 / 1,400
ESC	Generic 30 A BLHeli/SimonK, 5 V ~2 A BEC	Hobbywing Skywalker 30 A (or equal)	Continuous 30 A; 2 to 4S; 5 V BEC; adjustable low-voltage cutoff	1	700 / 1,500
Propeller	Generic 10×4.7 slow-fly plastic	APC 10×4.7 SF	Diameter 10 in (254 mm), pitch 4.7 in; carry spares	3 / 2 + 2 spare	300 / 700
Prop adapter / spinner	Collet adapter + rubber prop saver	Aluminium collet adapter + spinner	Bore must match the motor shaft	1	150 / 250
Battery	3S 2200 mAh 25C LiPo, XT60, generic	3S 2200 mAh 35 to 45C LiPo, branded	11.1 V nominal, 12.6 V full; about 185 g	1	1,900 / 3,000
Transmitter + receiver	FlySky FS-i6 + FS-iA6B	RadioMaster Pocket (ELRS) + ELRS PWM receiver	2.4 GHz; at least 4 channels; failsafe	1 set	4,800 / 12,000
Servos	4 × SG90 9 g micro (plastic gears)	4 × MG90S metal-gear micro	Torque about 1.6 vs 2.0 to 2.2 kg·cm at 4.8 to 6 V (typical); about 23 × 12 × 29 mm	4	500 / 1,200
Landing gear	None: hand launch, belly landing on a foam skid plate	2.5 mm piano-wire main gear, 2 × 55 mm foam wheels, tail skid	Optional, adds about 30 g	0 / 1 set	0 / 500
Connectors and wiring	XT60 × 2 pairs, 3.5 mm bullets × 3 pairs, 4 servo extensions (30 to 50 cm), 1 Y-lead, 1 m 16 AWG silicone wire	Same, higher-quality parts, 2 m wire, more heat-shrink	XT60 rated about 60 A (typical) against about 14 A needed	1 set	650 / 900
Adhesives and consumables	Hot glue, PVA (white glue), 5-min epoxy, foam-safe CA, tape, Velcro, rubber bands, heat-shrink, zip ties	Thin and medium CA, 5-min epoxy, PVA, sandpaper, Velcro, rubber bands	Choose glue that does not dissolve foam	1 set	1,150 / 1,200

Component	Recommendation 1 (budget)	Recommendation 2 (performance)	Key specifications	Qty	Cost ₹
Control and mounting hardware	4 nylon horns, 4 clevises, 1.2 mm wire pushrods, tape hinges, 3 mm ply firewall, M3 screws	CA hinges, carbon pushrods, same firewall	Horn height matches servo arm length	1 set	350 / 600
Support equipment (shared, optional)	2 to 3S balance charger, LiPo bag, watt meter, low-voltage alarm	Better charger with power supply, same others	Needed to charge safely and test thrust	1 set	2,300 / 4,100

3. Technical Justification (Part C)

3.1 Motor–propeller match and how to check the thrust

Why it matters: the motor and propeller must be matched. A prop that is too big overloads the motor and ESC (heat, burnt parts). One that is too small wastes the motor's power and gives too little thrust.

Step 1: decide how much thrust is needed. For a trainer-type aircraft with ground or hand launch, static thrust of about 0.6 to 0.8 times the weight is a common rule of thumb. Weight = 0.95 kg × 9.81 = 9.3 N, so we need 5.6 to 7.5 N (about 570 to 760 g). We aim for at least 7 N.

Step 2: fix the battery cell count first. KV (RPM per volt) and cell count together set the RPM. 2S is too weak for this mass. 4S would need a lower-KV, heavier motor. 3S is the common, cheap, well-supported choice, so 3S it is.

Step 3: choose KV and propeller together.

- Loaded RPM ≈ KV × voltage × 0.85 = 1000 × 11.1 × 0.85 ≈ **9,400 rpm** (0.85 is a typical loading factor).
- Pitch speed = pitch × RPM = 0.119 m × 9,400 / 60 ≈ **18.7 m/s**. This is about 1.5 times our 12 m/s cruise (a rule of thumb is 1.3 to 2 times), so the prop can still push at cruise and gives good thrust at low speed.
- Tip speed = π × 0.254 m × 9,400 / 60 ≈ 125 m/s, about Mach 0.37, well under the roughly Mach 0.7 where the efficiency drops.

Option	Static thrust	Top speed	Best for
Large, slow prop (10×4.7 on 1000 KV)	High per watt	Moderate	Takeoff, hand launch, slow flight
Small, fast prop (8×4.3 on 1400 KV)	Lower per watt	Higher	Speed models

A bigger propeller moves more air more slowly, which gives more thrust for the same power. That is why a slow, light trainer wants the large prop.

Step 4: estimate the thrust. Momentum theory: $T = (2\rho A \cdot P_{ideal}^2)^{(1/3)}$.

- Electrical power at full throttle = 11.1 V × 13 A ≈ 144 W (13 A is a typical datasheet value for this motor and prop, to be verified).
- Motor + ESC efficiency about 0.75, so shaft power ≈ 108 W.
- Figure of merit (how well the prop turns power into thrust) about 0.6, so P_{ideal} ≈ 65 W.
- Disk area $A = \pi \times (0.127\ m)^2 = 0.0507\ m^2$.
- $T = (2 \times 1.225 \times 0.0507 \times 65^2)^{(1/3)} \approx$ **8.1 N (about 820 g)**.

Manufacturer tables for this motor class with a 10×4.7 prop on 3S commonly show roughly 750 to 900 g, which agrees. We use a conservative **7.5 N (about 765 g)** as the design value, giving a thrust-to-weight ratio of 0.8.

Step 5: check available thrust by test.

1. Fix the plane firmly to a bench, with the propeller area clear of people and loose objects.
2. Put a watt meter between the battery and the ESC. Attach a spring (luggage) scale between the tail and a fixed anchor.
3. Run full throttle for about 5 seconds. Record peak thrust (scale), voltage and current (watt meter).
4. **Accept** if thrust is at least 7 N and current is no more than about 85% of the motor's rating.
5. If current is too high, use a smaller or lower-pitch prop (for example 9×4.7). If thrust is too low, check for battery voltage sag, a damaged prop or wrong motor direction.
6. Feel the motor and ESC after 30 seconds. They should be warm, not too hot to touch.

Why Recommendation 2 (SunnySky class)? It has the same KV and prop size, so the same calculation holds. You pay for better bearings, balance and quality control, which means less vibration (good for the airframe and electronics) and more consistent KV.

3.2 ESC rating and electrical margin

- **Expected maximum motor current:** about 13 A (Section 3.1).
- **Rule:** the ESC continuous rating should be at least 1.5 times the maximum motor current, and preferably 2 times for low-cost ESCs. That gives 20 to 26 A, so a **30 A ESC** is chosen. Margin = 30 / 13 ≈ **2.3 times**.
- **Why so much margin:** ESC ratings are measured with good cooling, and our ESC sits inside a foam fuselage. The motor current spikes when the propeller is damaged or the plane is held back. Cheap ESCs are often over-rated.
- **Voltage:** a 2 to 4S ESC covers 3S, whose full-charge voltage is 12.6 V.
- **BEC (the 5 V supply for the receiver and servos):** four micro servos draw about 0.3 to 0.6 A on average and about 2 A only if all stall together. A 2 A BEC is adequate. A linear BEC burns the extra voltage as heat: (12.6 − 5 V) × 0.5 A ≈ 3.8 W, which is warm, so give the ESC airflow. For bigger servos use a separate 5 V / 3 A switching BEC (UBEC).
- **Settings:** calibrate the throttle range, set brake off (so the prop windmills in a glide), timing medium, and low-voltage cutoff about 3.2 V per cell.
- **Rec 1 vs. Rec 2:** both meet the electrical numbers. The branded ESC gives more reliable ratings, protection and firmware, which is why it is the first upgrade I would pay for.

3.3 Battery choice and trade-offs

- **Chemistry:** LiPo gives high power for low weight. NiMH is heavier and weaker. Li-ion packs are light and long-lasting but can't deliver high current easily.
- **Cell count:** 3S, fixed by the motor and prop in Section 3.1.
- **Capacity:** with an average current of about 8 A (a mix of climbs and cruise) and 80% usable capacity (never fully drain LiPo), 2.2 Ah × 0.8 = 1.76 Ah, so flight time ≈ 1.76 / 8 = 0.22 h ≈ **13 minutes**. This meets the 10-minute target with margin.
- **Discharge rating (C):** required C = peak current / capacity = 13 / 2.2 ≈ 6 C. Labels are often inflated, so we choose at least 20 to 25C. A generic 25C pack gives about 55 A on paper, 4 times the need.
- **Capacity trade-off:**

Pack	Mass	Flight time (est.)	Effect
3S 1500 mAh	about 130 g	about 8 min	Lighter, but short flights
3S 2200 mAh	**about 185 g**	**about 13 min**	**Best balance**
3S 3000 mAh	about 250 g	about 18 min	About 7% more total mass, so stall speed about 3% higher

- **Rec 1 vs. Rec 2:** the branded pack is more likely to deliver its rated current, hold its capacity, and not puff up. **Never save money on the battery**: a bad LiPo is the largest fire risk in the build.
- **Safety:** charge only with a balance charger inside a LiPo bag, never unattended. Store at about 3.8 V per cell. Stop flying at about 3.5 V per cell under load. Do not use damaged or swollen packs.